

CRITICAL PIPELINE INTEGRITY ASSESSMENT REPORT

Client: National Hydrocarbon Transport Corp.
Pipeline Section: Sector-7 Western Corridor
Pipeline ID: NHTC-WC-007
Inspection Agency: Advanced Integrity Consultants
Inspection Date: 15 June 2026
Report Number: AIC/CRI/2026/7781

Executive Summary

The pipeline section is assessed to be in POOR condition. Multiple integrity threats have been identified requiring immediate corrective action.

Cathodic Protection Survey (PSP)

Average PSP Reading: -0.71 V
Minimum PSP Reading: -0.63 V
Survey Coverage: 62%

DCVG Survey

Total Coating Defects Identified: 128
Severe Defects: 42
Moderate Defects: 51
Minor Defects: 35
Average DCVG Severity: 68.4%

Corrosion Coupon Analysis

Corrosion Rate: 0.82 mm/year
Iron Content: 480 ppm
Sulfide Presence: High

Corrosion Probe Monitoring

Average Corrosion Rate: 0.73 mm/year
Peak Corrosion Rate: 1.15 mm/year

Soil Resistivity Assessment

Average Soil Resistivity: 620 ohm-cm
Lowest Recorded Resistivity: 410 ohm-cm

AC Interference Study

Average AC Voltage: 41.7 V
Peak AC Voltage: 67.3 V

ILI Summary

Metal Loss Features: 287
High Priority Features: 46

Critical Features: 12
Dent Features: 58
Crack Indications: 9

Leak Detection System Test

Static Leak Simulation Detection Time: 38.4 min
Dynamic Leak Simulation Detection Time: 51.8 min
Detection Success Rate: 54%

Compliance & Maintenance Data

Scraper Pigging Compliance: 41%
Pig Residue Analysis Completed: No
Inspection Schedule Compliance: 48%
Audit Compliance Score: 37%
Right of Way Compliance: 52%
Data Completeness: 61%

Engineer Remarks

Immediate integrity intervention is recommended. Significant coating degradation, high corrosion activity, inadequate cathodic protection, elevated AC interference, and multiple critical ILI anomalies present an increased probability of failure.